ORFE 409 Midterm Examination (4 pages)

Due by 12 noon on Tuesday, October 21, 2025

There are a total of 5 problems.

Please follow the honor code and let this represent your own work.

All simulation code should be based solely on the use of simulation packages for standard uniform random variables.

Problem 1 (20 points, continued on the next page): Consider the $M/G/1/\infty$ queueing system with the following three i.i.d. service time random variables (with mean one):

1. **Erlang-2 (E_2):** This is a Gamma$(2.0, 2)$ random variable.
2. **Exponential (M):** This is an Exponential(1.0) random variable.
3. **Hyperexponential-2 (H_2):** We choose this one to be the selection of a Exponential(1/9) random variable (which has mean 9.0) with probability 0.1 *or* an Exponential(9.0) random variable (which has mean 1/9) with probability 0.9.

Consider the $M/G/1/\infty$ queueing system with these three i.i.d. service time random variables. Moreover, we consider the following two service disciplines for this queueing system:

1. **FIFO:** This is the *first-in, first-out* service discipline. The server processes the first job at the head of the line at a unit rate.
2. **PS:** This is the *processor sharing* service discipline. The size of the job is the time that it would take for the server to process it if there were no other jobs. As long as there are n jobs in the system, the processing time will be n times longer. This is equivalent to saying that the processing rate is n times slower.

Create the following five graphs. Given a mean Poisson arrival rate of $\lambda = 0.9$ and initial queue length $Q(0) = 0$. The first three graphs are two plots each and the last two graphs are three plots each of the mean number $\mathrm{E}[Q(t)]$ in the system (the number of customers in the queue or buffer plus the customer in service) as a function of time t (let t range from 0 to 600, where $\mathrm{E}[Q(t)]$ is computed at each time point $k * 0.2$, where $k = 0, 1, \ldots, 3000$:

1. The service time distribution M is held fixed but we vary the service discipline.

2. The service time distribution E_2 is held fixed but we vary the service discipline.

3. The service time distribution H_2 is held fixed but we vary the service discipline.

4. The service discipline *FIFO* is held fixed but we vary the service time distribution.

5. The service discipline *PS* is held fixed but we vary the service time distribution.

Continuation of Problem 1: Please title each graph, label the x-axis and y-axis, and create a legend box to distinguish each of the three plots. [Note: You need only stop the simulation at time $t = 600$. You do not have to serve the customers until the last one leaves.]

Problem 2 (10 points): Let X_n be a negative binomial random variable with n degrees of freedom or Binomial$(p, -n)$ with $0 < p < 1$.

1. Show that for all $n \geq 2$, we have

$$\frac{n}{n + X_n} \geq \frac{n-1}{n-1+X_n} \quad \text{and} \quad \mathrm{E}\left[\frac{n-1}{n-1+X_n}\right] = 1 - p.$$

2. Now show how X_n can be simulated such that

$$\lim_{n\to\infty} \frac{n}{n + X_n} = \lim_{n\to\infty} \frac{n-1}{n-1+X_n} = 1 - p.$$

Problem 3 (10 points): Given some real number $\alpha > 1$, define $\log_\alpha x$ to be the *logarithm base* α of x or the inverse of function α^x (Note: When we set the base to be $\alpha = e = 2.71828\ldots$, then we just use log). Consider $0 < p < 1$ and some standard uniform random variable $\mathcal{U}$.

1. Show that $\lfloor \log_{1/p}(1/\mathcal{U}) \rfloor$ is a Geometric(p) random variable (as defined in the class lecture notes), where $\lfloor x \rfloor$ or the *floor* function of x is the largest integer *less* than or equal to x.

2. Combine this result with Problem 2 to simulate a table of 3 columns and 5 rows where a given row is

$$n, \quad \frac{n}{n + X_n}, \quad \frac{n-1}{n-1+X_n}$$

for $p = 0.3$ and $n = 2, 4, 8, 16, 32$. Finally, recreate this table two more times and display all three tables.

Problem 4 (20 points): Let the price of a stock at time $t > 0$ be $S(t)$, where μ guaranteed yield, r is the interest rate, σ is the volatility, $S(0)$ is the present value of the stock, and

$$S(t) = S(0) \cdot \exp\left(\sigma W(t) + (\mu - \sigma^2/2)t\right)$$

where $\{W(t)\,|0 \le t \le T\}$ is standard Brownian motion. The general theory for complete markets says that if X is measurable with respect to $\{W(t)\,|0 \le t \le T\}$ (or equivalently $\{S(t)\,|0 \le t \le T\}$) and if V equals the fair market value for an option today ($t = 0$) given that it has value X when exercised at time T, then

$$V = \mathrm{E}\left[X \cdot \exp\left(\left(\frac{r-\mu}{\sigma}\right) W(T) - \left(r + \frac{(\mu - r)^2}{2\sigma^2}\right) T\right)\right]$$

Now we specifically describe three types of options, where K is always the strike price:

1. **European Option:** The value of the option is a function of the price for the underlying stock at time T or
$$X = (S(T) - K)^+ .$$
For this case, V is the Black-Scholes formula or
$$V = S(0) \cdot \Phi\left(\frac{\log(S(0)/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}\right) - K\, e^{-rT} \cdot \Phi\left(\frac{\log(S(0)/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}\right),$$
where
$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-y^2/2}\, dy.$$

2. **Asian Option:** The value of the option is a function of the price for the underlying stock averaged over the interval $[0, T]$ or
$$X = \left(\frac{1}{T}\int_0^T S(u)\, du - K\right)^+ .$$

3. **Look-Back Option:** The value of the option is a function of the price for the underlying stock maximized over the interval $[0, T]$ or
$$X = \left(\max_{0 \le u \le T} S(u) - K\right)^+ .$$

Using simulation, compute these three options as a function of T, where $S(0) = 59$, $K = 60$, $\mu = 0.1$, $r = 0.05$, $\sigma = 0.35$ and

$$T = 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5.$$

Plot all three curves for the simulated stock option prices on the same graph. Also add a fourth plot of the Black Scholes formula for the same time points. Please title each graph, label the x-axis and y-axis, and create a legend box to distinguish each of the four plots. [Hint: Look at the definition of Brownian motion in the class notes and note that $\sqrt{\Delta t} \cdot \mathfrak{G}$, where the second factor $\mathfrak{G}$ is a standard Gaussian random variable, always simulates $W(t + \Delta t) - W(t)$ for all t. Also, a reasonable size for Δt is of the order 10^{-4} to 10^{-6}].

Problem 5 (20 points): Now simulate an M/M/1/k queue by using the uniformization method as discussed in lecture notes for this week (6b) where we set $k = 20$.

Its state space is $\{0, 1, \ldots, 20\}$ and its potential transition rates are:

$$a(i,j) = \begin{cases} \lambda & \text{for all integers } i = 0, \ldots, 19 \text{ and } j = i+1, \\ \mu & \text{for all integers } i = 1, \ldots, 20 \text{ and } j = i-1, \\ 0 & \text{otherwise.} \end{cases}$$

We let the resulting queueing process $\{Q(t) \,|\, t \geq 0\}$ evolve but stop it at the *random* time T where

$$T = \min \{t \,|\, Q(t) = 0 \text{ or } Q(t) = k\}.$$

Hence T is the first time the queueing system reaches state 0 or 20.

We always assume that $\mu = 1$. Now do the following:

1. Use simulation to estimate the mean value of this random time or ET for the following 5 initial load cases:

   $$Q(0) = 2, 6, 10, 14, 18$$

   as well as

   $$\lambda = 0.4, 1, 2.5.$$

   This results in a total of 15 pairs of initial load and arrival rate values.

2. Now for a single arrival rate value, plot (x, y) where x equals every initial load value and y equals the corresponding mean of the random time T. You can also include the points $(0, 0)$ and $(20, 0)$. Connect every two consecutive points by a line segment for a continuous, piece-wise linear curve. You can place all three plots on the same graph. Please label each plot.